

R.M.N.N.Rathnayaka

0757105455 | nipunnirman999@gmail.com | LinkedIn : <https://www.linkedin.com/in/nipun-rathnayaka-889714250/> | <https://nipun-nirman.vercel.app>
(GitHub- <https://github.com/nipunnirman>) | Colombo , Sri Lanka



PROFESSIONAL SUMMARY

- Passionate and motivated Computer Science undergraduate aiming to become a skilled **Machine Learning Engineer** and **Full-Stack Developer**. Equipped with strong programming fundamentals, cloud knowledge, and hands-on experience in both AI and web development. Eager to build intelligent, scalable applications that solve real-world problems. Actively seeking opportunities to grow through collaborative projects, internships, and continuous learning.

SKILLS

- **Programming Languages:** Python, Java, JavaScript, HTML, CSS, PHP
- **Machine learning:** Scikit-learn ,TensorFlow,Pandas,NumPy,Decision Trees ,LLMs
- **Libraries/Frameworks:** Pandas, React
- **Tools & Platforms:** Git, Docker (learning), Azure (basics)
- **Concepts:** Cloud Computing, Data Analysis, Problem Solving
- **Databases :** MySQL,MongoDB ,ChromaDB
- **Cloud & CI/CD:**Azure ,Github Actions , MLflow

A/L BIOLOGY AI QUESTION ANSWER ASSISTANT

- Built a multi agent RAG system using a 3 agent LangGraph pipeline (Retrieval, Summarization, Verification) to answer A/L Biology questions from Grade 12 & 13 resource books with chunk level citations and hallucination detection.
- Indexed Biology textbook PDFs into Pinecone vector database and implemented Markdown formatted responses including comparison tables, structured lists, and bold key terms rendered via React frontend.
- Deployed full stack application on Vercel with Python serverless backend (FastAPI) and Vite built React chatbot UI featuring responsive design and real-time typing indicators.
- Technologies:python,FastAPI,LangChain,LangGraph,Pinecorn,React,Vercel

Multi agent RAG System Using NASA data

- Developed an advanced Retrieval-Augmented Generation (RAG) system combining vector search (ChromaDB) and keyword search (BM25) for semantic and lexical retrieval of NASA technical documents, achieving enhanced accuracy through reciprocal rank fusion.
- Deployed interactive chatbot interface using Streamlit frontend and FastAPI backend, containerized with Docker for scalable deployment on AWS Elastic Beanstalk.
- Technologies: [python](#),[Langchain](#) , [ChromaDB](#),[OpenAI](#),[Sentence Transformers](#),[Fast API](#) ,[Docker](#)

Dementia Risk Prediction Model

- Developed a binary classification model achieving 94.19% ROC-AUC for dementia prediction using 195,196 clinical records from the NACC dataset with minimal overfitting (0.56% gap).
- Implemented comprehensive ML pipeline across 5 Jupyter notebooks covering data cleaning, feature engineering (18 derived features), and model training with XGBoost, Random Forest, and Logistic Regression.
- Conducted rigorous validation using 5-fold cross-validation and eliminated data leakage through systematic removal of diagnosis-related features, achieving 90.16% accuracy with 83.27% F1-score.
- Technologies: [python](#),[XGboost](#),[Scikit-Learn](#),[Pandas](#),[Numpy](#),[Matplotlib](#)

Exoplanet Discovery using Kepler Data (Machine Learning + Streamlit)

- Developed a binary classification model using XGBoost to predict exoplanet candidates from NASA Kepler telescope data, distinguishing real exoplanets from false positives using 15 astrophysical features including orbital period, transit depth, and stellar properties.

- Built an interactive Streamlit web application enabling real-time exoplanet probability predictions with SHAP explainability integration for interpretable feature impact visualization and scientific analysis.
- Implemented comprehensive data preprocessing pipeline with scientific scaling and validation using ROCAUC and PR-AUC metrics on the Kepler Objects of Interest (KOI) catalog dataset.
- Technologies: [python](#),[XGboost](#),[Scikit-Learn](#),[Pandas](#),[Numpy](#)

Cricket Toss Analysis & Win Prediction System - Sri Lankan Grounds

- Developed a comprehensive cricket analytics platform analyzing 50+ matches across 7 Sri Lankan cricket grounds, combining toss decisions, player performance, and weather data to predict match outcomes with 75%+ accuracy and winning scores .
- Built machine learning pipeline using XGBoost and Random Forest for three predictive models: optimal toss decision classifier, match winner predictor, and score regression model, with feature engineering incorporating ground characteristics, team form, and meteorological data. • Technologies: [python](#),[XGboost](#),[Scikit-Learn](#),[Pandas](#),[Numpy](#),[Matplotlib](#) ,[Streamlit](#) ,[Plotly](#),[FastAPI](#) ,[Docker](#) ,[OpenWeatherMap API](#)

Enterprise Chat Session Feedback System

- Developed a multi-tenant full-stack SaaS platform for dynamic post-chat feedback collection, effectively eliminating Insecure Direct Object Reference (IDOR) vulnerabilities via secure UUID tokenization.
- Implemented decoupled frontend-backend architecture featuring 2 distinct UI portals (Admin and Customer) built with React, integrating with strict REST API endpoints managed by a Spring Boot service layer.
- Engineered a high-performance MongoDB data layer utilizing an Upsert pattern for real-time form configuration, establishing zero-redundancy state management (PENDING, RESPONDED, EXPIRED) with strict 24-hour expiration validation.
- Technologies: Kotlin, Spring Boot, React, Vite, MongoDB, REST API

LICENCES AND CERTIFICATIONS

- AI Engineer-StemLink
- Microsoft Azure AI-900
- Python – University of Moratuwa
- LinkedIn Learning – Python and Java
- Deep Learning With Python(LinkedIn Learning)
- Machine Learning (Self-paced course)
- Apache DAG Airflow3
- LangGraph Essential – Python
- NASA Space Apps Challenge : Hunting for Exoplanets with AI

EDUCATION

- University of Westminster BSc(Hons) in Computer Science 2025 – 2028 (Expected)
- GCE Advanced level in Maths stream (completed)

VOLUNTEERING

- Volunteering at SASNAKA SANSADA GANITHA SAVIYA

REFEREES

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|---|---|
| <ul style="list-style-type: none">• Sudesh Rathnayake
Tech Lead
Impact.com
Email: sudesh.rathnayake-mudiyanselage@impact.com
Phone: +1 (431) 338 5060 | <ul style="list-style-type: none">• Mr.G.A.Pubudu Janith
Lecturer
Infomatics Institute of Technology
Email: Pubudu.a@iit.ac.lk
Phone: +94 77 127 4857 |
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